

Week 2 demo — fallback captures

City bike-sharing: one architect-critic cycle on a requirements document

Captured with Claude Code, Sonnet 5 (low effort), September 2026

When to use this deck

The live AI is not answering, so here is what the same prompts produced in a real run. We read and critique this requirements document exactly as we would have done live.

Prompt #1

“Generate a requirements document for a city bike-sharing app: users rent and return bicycles at stations across a city; payment is by app; staff rebalance bikes between stations. Cover functional, non-functional, and domain requirements.”

AI's answer — scope and features

City Bike-Sharing App: Requirements Document

Stakeholders: Riders (registered and occasional), rebalancing and maintenance staff, operations managers, finance, city authority, and payment providers.

[...]

- ▶ FR-2: Users can log in, log out, and reset their password. Optional social or biometric login.
- ▶ FR-9: Users can reserve a bike for up to 10 minutes.
- ▶ FR-14: Users can pause a rental by locking the bike temporarily, with the time billed.
- ▶ FR-15: Each user may have one active rental by default. Group rentals of several bikes under one account are a configurable option.
- ▶ FR-21: The system supports pay-per-ride, day passes, and monthly or annual memberships.
- ▶ FR-25: The system supports promo codes and discounts.
- ▶ FR-29: The system forecasts demand from history, time of day, weather, and events to prioritize tasks.
- ▶ FR-37: Administrators can export data to CSV and to an API for the city authority.
- ▶ FR-38: The system sends push, email, or SMS notifications for receipts, long rentals, payment failures, reservation expiry, and service alerts.

AI's answer — non-functional and domain

- ▶ NFR-4: The back end is available 99.9% of the time, excluding planned maintenance.
- ▶ NFR-8: The system supports at least 100,000 registered users, 10,000 concurrent rentals, and 1,000 stations, and can scale horizontally beyond that.
- ▶ NFR-9: All traffic uses TLS 1.2 or later. Data at rest is encrypted.
- ▶ NFR-10: The system never stores card data itself. It uses tokenization by the payment provider (PCI DSS).
- ▶ NFR-18: The app meets WCAG 2.2 AA. It supports screen readers and large text.
- ▶ NFR-19: The app is available in the city's main languages and supports localized currency and date formats.
- ▶ NFR-22: The system has automated tests with at least 80% coverage of core rental and payment logic.

[...]

AI's answer — are e-bikes in scope?

- ▶ FR-7: Users can search and filter stations, for example by e-bikes or by nearest.

[...]

- ▶ DR-13: Bikes carry GPS or telemetry to locate them. E-bikes report battery level, and low-battery bikes are not offered for rent.

[...]

Open questions:

1. Are e-bikes, dockless bikes, or scooters in scope?

[...]

Prompt #2

"Rewrite the requirements organized by use case. Each use case is one user goal from my prompt — renting, returning, paying, rebalancing; nothing else. For each: who does it, what's the success criterion, and one realistic non-functional constraint. Use only the roles I named. Drop technology and standards choices — those are implementation, not requirements. Don't invent numbers: if a threshold needs a decision from me, write TBD."

Al's revised answer — by use case (1/2)

Roles: **user** and **staff**. Thresholds marked TBD need your decision.

UC-1: Rent a bicycle

- ▶ **Actor:** User.
- ▶ **Goal:** Take a bicycle from a station for a ride.
- ▶ **Success criterion:** The user picks up a bicycle at a station and it is released to them. The system records the rental as started at that moment and associates it with the user.
- ▶ **Non-functional constraint (performance):** The time from the user's request to bicycle release is at most TBD seconds under normal conditions.

UC-2: Return a bicycle

- ▶ **Actor:** User.
- ▶ **Goal:** End a rental by leaving the bicycle at a station.
- ▶ **Success criterion:** The user leaves the bicycle at a station with a free space. The system records the rental as ended, makes the bicycle available at that station, and shows the user a confirmation.
- ▶ **Non-functional constraint (reliability):** A return is never lost or double-counted if the connection drops or the user retries. The rental ends exactly once.

AI's revised answer — by use case (2/2)

UC-3: Pay for a rental

- ▶ **Actor:** User.
- ▶ **Goal:** Pay for a completed rental through the app.
- ▶ **Success criterion:** The user is charged the correct amount for the rental, and the payment is recorded against that rental. The user can see a receipt in the app.
- ▶ **Non-functional constraint (security and privacy):** Only the user's own payment details are used, and they are protected from disclosure to other users and to staff.

UC-4: Rebalance bicycles between stations

- ▶ **Actor:** Staff.
- ▶ **Goal:** Move bicycles from stations with too many to stations with too few.
- ▶ **Success criterion:** After a move, the source and destination stations show the correct bicycle counts. Moved bicycles are available for rent at the destination, and no station exceeds its capacity.
- ▶ **Non-functional constraint (data accuracy):** Station bicycle counts shown to users and staff are no more than TBD seconds out of date.

Reserve prompt — if the first document is too clean

“Now redo this assuming the city has 50,000 daily riders and 2,000 bikes across 200 stations.”

Reserve answer — it did the arithmetic

The fleet size doesn't fit the ridership. 50,000 riders a day on 2,000 bikes means about 25 trips per bike per day. Busy systems usually see 4 to 10. Average stations hold only 10 bikes ($2,000 \div 200$). I've written the requirements to fit that load. Please confirm whether "50,000 daily riders" means 50,000 trips or unique users, and whether the fleet is really 2,000 bikes.

Sizing assumptions [...]

Item	Value	Basis
Peak hour	~10 to 12% of daily rides, or 5,000 to 9,000 rides/hour	Commute peaks
Peak unlock rate	~2.5 per second average, ~10 per second burst	Bunching at station clusters
Docks	~3,600 ($200 \text{ stations} \times \sim 18 \text{ docks}$)	~1.8 docks per bike
Registered users	~300,000 to 500,000	Assumed, for capacity planning

Reserve answer — guesses become requirements

- ▶ FR-D2: Show availability trends and a short-term prediction (“likely empty in 10 min”), since stations hold only about 10 bikes.
- ▶ FR-D5: Reserve a bike for up to 10 minutes, one reservation at a time. Reservations may not hold more than 20% of a station’s bikes, so they can’t starve walk-up riders.

[...]

ID	Category	Requirement
NFR-2	Through-put	Sustain 10 unlocks/s and 10 returns/s. Handle ~15,000 telemetry messages/minute. Design for 3x headroom for growth or events.
NFR-3	Capacity	500,000 registered users, 30,000 concurrent app sessions at peak, and 3,600 connected docks and 2,000 bikes.

[...]

- ▶ DR-14: Comply with the city’s operating permit, including sharing anonymized trip data (for example, in GBFS format).
- ▶ DR-19: Target service levels are stations empty less than 5% of peak time and full less than 5% of peak time. These need confirmation with the city.